MIL-E-62462B(AT)
26 August 1985
SUPERSEDING
MIL-E-62462A(AT)
AMENDMENT 1
29 June 1984

MILITARY SPECIFICATION

ENGINE, DIESEL, LIQUID-COOLED, 8 CYLINDER; 4 CYCLE, V-TYPE, 295 H.P. (220 KW)

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirement for a liquid cooled, 4 cycle, 90° vee, 8 cylinder internal combustion, compression-ignition engine. At 2600 revolutions per minute (rpm), the diesel engine is rated at 295 horsepower (hp) [220 kilowatts (kw)] (see 6.4)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2815

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SPECIFICATIONS
FEDERAL

O-A-548	- Antifreeze/Coolant, Engine, Ethylene, Glycol, Inhibited, Concentrated.
O-I-490	- Inhibitor, Corrosion, Liquid Cooling System.
P-C-437	- Cleaning Compound High Pressure (Steam) Cleaner.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-P-514	- Plates Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Heavy Duty.
MIL-A-11755	- Antifreeze, Arctic Type.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-193	- Paint Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics Measurement of.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-45662	- Calibration System Requirements.
MS35844	- Plug, Machine Thread, Magnetic, Drain.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

12325839	- Engine Cummins V903.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J1349 - Engine Test Code - Spark Ignition and Diesel.

(Applications for copies of SAE publications should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pensylvania, 15096.)

NATO INTERNATIONAL STAFF - DEFENSE SUPPORT DIVISION

AEP-5 (Jan 84) - NATO Standard Diesel and Spark Ignition Engines Laboratory Test.

(Application for copies of NATO publications should be addressed to NATO, Military Agency for Standardization (MAS), 35 Chesham Place, London SWI, England.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish diesel engines which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent diesel engines delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Design and construction.

3.2.1 Material, parts, and processes. Materials shall be as specified herein and in referenced specifications, standards, and drawiings. Engine construction and assembly shall be in accordance with drawing 12325838.

3.2.2 Weight. The dry weight of the engine as shown on drawing 12325839 shall be not more than 2350 pounds (lb) [1066 kilograms (kg)].

3.2.3 Lifting eyes. The engine shall be equipped with lifting eyes as shown on drawing 12325839.

3.2.4 Diagnostic test points. The engine shall be provided with accessible test points for temperature and pressure measurements in the air, fuel, cooling and lubrication systems and RPM measurement as shown on Drawing 12325839.

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3.2.5 Starter. The engine starter shall conform to system requirements, and shall operate at nominal voltage of 24 volts direct current (Vdc).

3.2.6 Drain plugs. Engine oil pan shall contain a magnetic plug conforming to MS35844.

3.2.7 Oil level. Permanent markings shall be provided on the dipstick to indicate quantity of oil to be added when oil level has dropped to add mark.

3.2.8 Interchangeability. To provide for interchangeability, the engine including accessories, shall conform to this specification and to drawing 12325839. Any changes affecting physical, functional or electrical interfaces shall require prior approval by the acquisitioning activity. All subassemblies having the same part number shall be functionally and dimensionally interchangeable.

3.3 Performance.

3.3.1 Engine performance. The following engine performance characteristics shall conform to figure 1 for those curves labeled "maximum". Performance shall be obtained using fuel conforming to VV-F-800, grade 2 (see 4.6.1.1).

a. Speed range
b. Brake horsepower (kilowatts)
c. Torque
d. Fuel consumption
e. Governed speed

3.3.1.1 Idle speed. The engine shall maintain a satisfactory idle speed of 800 ± 25 rpm (see 4.6.1.2).

3.3.1.2 Maximum governed speed. The maximum no load governed speed shall be 2965 rpm (see 4.6.1.3). The maximum full load governed speed shall be 2600 rpm.

3.3.1.3 Exhaust smoke density. With the engine operating at full load at speeds of 1800 and 2600 rpm, the exhaust smoke density shall not exceed the limits specified in table I. As an option, percent of opacity method may be used with limits as specified in table I (see 4.6.1.4).

TABLE I. Exhaust smoke density.

Engine speed (rpm)	Bosch rating	Opacity limit (max)
1800 ± 100	3.7	12%
2600 ± 100	3.2	9%

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3.3.1.4 Heat rejection. Total engine heat rejection to coolant shall be not more than 8400 British Thermal units per minute (BTU/minute) [8862 kilojoules per minute (kJ/minute)] under the following conditions (see 4.6.1.5):

a. Engine speed, 2600 rpm.
b. Engine air flow, 610 cubic feet per minute (cfm) [0.288 cubic meters per second (m^3/s)] at the inlet.
c. Coolant flow, 127 gallons per minute (gpm) [481 liters per minute (L/m)].
d. Torque, 596 pound feet (lb-ft) [808 Newton-meter (N m)].
e. Coolant temperature, 185 degrees Fahrenheit (°F) [85 degrees Celsius (°C)].

3.3.1.5 Engine starting. In an ambient temperature range of plus 120°F (49°C) to minus 25°F (-32°C), the engine shall be capable of being started in one minute or less (see 4.6.1.6). Ether starting fluid may be used at ambient temperatures below 32°F (0°C).

3.3.1.6 Coolant. Engine shall perform as specified herein using water and antifreeze compound conforming to O-A-548, arctic-type antifreeze compound conforming to MIL-C-11755, or water and corrosion inhibitor conforming to O-I-490 (see 4.6.1.7).

3.3.1.7 Normal operating temperatures. The engine shall be capable of operating at the following temperatures without abnormal engine degradation (see 4.6.1.7).

a. Oil sump: Minimum 140°F (60°C) and maximum 275°F (135°C) (measured at drain plug)

b. Coolant: Maximum 230°F (110°C) (measured at the inlet to the transmission oil cooler)

3.3.1.8 Air inlet and exhaust restrictions. The engine shall perform as specified herein at full rated power and speed with the following maximum flow restrictions (see 4.6.1.7):

a. Air inlet restriction: 25 inches of water (in. H_2O) [635 millimeters of water (mm H_2O)].

b. Exhaust restriction: 3.0 inches of mercury (in. Hg) (76 mm H_2O).

3.3.1.9 Oil. The engine shall perform as specified herein using oil conforming to MIL-L-2104 for an ambient temperature range of minus 10°F (-12°C) to plus 120°F (49°C) and oil conforming to MIL-L-46167 for arctic operation [0°F (-18°C) to -65°F (-54°C)] (see 4.6.1.7).

3.3.1.10 Fuel inlet restriction. The engine shall perform as specified herein at full rated power and rated load with a maximum fuel inlet restriction of (see 4.6.1.7):

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4.0 in. Hg (102 mm Hg) (clean filter element)

8.0 in. Hg (203 mm Hg) (dirty filter element)

3.3.1.10.1 Fuel. The engine shall perform as specified herein use DF-2 grade fuel in accordance with VV-F-800. DF-1 or DF-2 grades shall be required for low temperature ambient conditions as specified (see 4.6.1.6).

3.3.1.11 Slope operation. The engine must start and be capable of operating with the performance characteristics specified in 3.3.1 at a 70 percent front to rear or rear to front slope and at a 45 percent side slope (either direction). The engine shall be capable of operating on those slopes when the oil level on the dipstick is at the "add" mark (determined on level ground). There shall be no evidence of faulty lubrication, coolant or fuel leakage or other malfunction (see 4.6.1.9).

3.3.1.12 Endurance. The engine shall be capable of passing AEP-5 (revised January 1984) NATO standard engine 400 hour endurance laboratory test. The 400 hour test is divided into four periods of 100 hours each with each 100 hours divided into ten, ten hour schedules as defined in the AEP-5 document (see 4.6.1.10).

3.3.2 Steam and waterjet cleaning. The engine shall be capable of withstanding steam cleaning compound conforming to P-C-437 and waterjet washing without deterioration or leakage of seals, gaskets or electical components. Paint removal shall not be considered a deficiency (see 4.6.2).

3.3.3 Leakage. When the engine is being operated for any or all tests specified in this specification, there shall be no evidence of "leaks" or "drips" in the oil, fuel, or coolant circuits (see 4.6.3 and 6.3). A "weep" or "seep" is permissable, but shall be recorded.

3.3.4 Environmental conditions. The engine shall meet all performance requirements before, during, and immediately after the environmental conditions specified herein and in accordance with general and specific requirements of MIL-STD-810 (see 4.6.4).

3.3.5 Electromagnetic radiation. Electromagnetic emissions from the engine shall be in accordance with the requirements of MIL-STD-461 (see 4.6.5).

3.4 Protective coating. All exposed exterior surfaces of the engine shall be finished per MIL-STD-193 (see 4.7 and 6.5).

3.5 Identification and marking. The engine shall be identified in accordance with MIL-STD-130 (see 4.8).

3.5.1 Nameplate. Unless otherwise specified on the applicable drawing, the nameplate, data, and instruction plates shall conform to the applicable provisions of MIL-P-514 (see 4.8.1).

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3.5.2 Accessories and equipment. All accessories and equipment shall be identified in accordance with auxiliary manufacturers standard practice and the I.D. markings shall include the manufacturer's part number, serial number if available, and date of manufacture.

3.6 Workmanship. The workmanship shall be of such quality as to assure that engines procured under contract meet all requirements specified herein. The engines produced shall be free of any defects or safety hazards which affect its performance or serviceability (see 4.9).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials and construction. To determine conformance to 3.2, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Acceptance tests (see 4.5.3.1).
    3. Control tests (see 4.5.3.2).

4.3 Test conditions. Unless otherwise specified, all tests shall be conducted at 70 $\pm$ 10°F.

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4.4 First article inspection. First article inspections are categorized as preproduction inspection and initial production inspection. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply diesel engines that are fully representative of those inspected as a first article sample. Any changes or deviation that affect form, fit or function of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two diesel engines. Preproduction inspection shall consist of examination as specified in table II and tests specified in table III.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two units, from the first ten engines produced under the production contract for initial production inspection. Initial production units shall be examined as specified in table II and tested as specified in table III.

4.4.3 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval and accept production units until corrective action by the contractor has been approved by the Government.

NOTE: Assemblies subjected to first article endurance tests shall not be delivered to the Government, but will be retained and indelibly marked, DO NOT USE.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the diesel engines from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

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TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0 % Defective	
101	Incomplete assembly.	Visual
102	Valve tappet clearance (each valve): intake and exhaust valves improperly adjusted.	Visual and functional
103	Fuel, oil and coolant leaks: leakage (see 6.3).	Visual
104	Governor: malfunction; not sealed.	Visual and functional
105	Improper torque on accessible cylinder head bolts, intake and exhaust manifold flange bolts that can be reached without breaking seals or removing cover.	Certified torque equipment
106	Crankshaft: excessive or restrictive end play.	Visual and functional
107	Fuel system components: malfunction; damage; leaks (see 6.3).	Visual and functional
108	Fuel injection system timing and components: malfunction; improper adjustment.	Visual and functional
Minor	AQL 2.5% Defective	
201	Fuel lines: damaged, rubbing, improper support.	Visual
202	Oil dipstick gage: missing.	Visual
203	Oil level: overfilled, below safe operating level.	Visual
204	Bolt on components loose; nuts or bolts missing, heads rounded or deformed.	Visual/ tactile
205	Threaded holes: correct size, proper depth.	Go and not go thread gages
206	Engine data plate: missing.	Visual
207	Painting: sags, runs, inadequate or improper coverage, none in threaded holes.	Visual

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent inspection). Each engine assembly shall be subjected to the tests specified in table III.

4.5.3.2 Control tests. Control tests shall be conducted on one engine assembly from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The engine assemblies shall be subjected to the tests specified in table III.

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4.5.4 Failure. Failure of any engine assembly to pass any of the specified tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

NOTE: Any engine assembly subjected to preproduction, initial production, or endurance tests shall not be delivered to the Government as an end item of the contract until the engine has been disassembled and examined for wear and damage. Wear limits shall not exceed those specified for overhaul by the manufacturer. Scuffing, galling, or burning of parts or surfaces shall not be permitted. Subsequent to the inspection performed above, the engine may be reassembled using all parts that meet new part drawing requirements, or replacing worn or damaged parts with new parts. Following reassembly the engine shall pass the break-in-run and acceptance tests as specified herein.

4.6 Methods of tests. Unless otherwise specified all tests specified herein shall be conducted on an engine dynamometer.

4.6.1 Engine performance and endurance.

4.6.1.1 Engine performance. To determine conformance to 3.3.1, verify that the engine performance capabilities conform to the maximum curves specified in figure 1.

4.6.1.2 Idle speed. To determine conformance to 3.3.1.1, verify that the idle speed requirement of 800 $\pm$ 25 rpm is obtained.

4.6.1.3 Maximum governed speed. To determine conformance to 3.3.1.2, verify that the no load governed speed is not more than 2965 rpm and the full load governed speed is not more than 2600 rpm.

4.6.1.4 Exhaust smoke density. Verify that the exhaust smoke density meets the requirements of 3.3.1.3 and table I.

4.6.1.5 Heat rejection. Verify that the total engine heat rejection to coolant does not exceed the requirement of 3.3.1.4.

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TABLE III. Classification of tests.

Title	Requirement	Test	First article	Quality conformance	
				Acceptance (100%)	Control
Engine performance	3.3.1	4.6.1.1	X	X	
Idle speed	3.3.1.1	4.6.1.2	X	X	
Maximum governed speed	3.3.1.2	4.6.1.3	X	X	
Exhaust smoke density	3.3.1.3	4.6.1.4	X	X	
Heat rejection	3.3.1.4	4.6.1.5	X		X
Engine starting	3.3.1.5	4.6.1.6	X		X
Operating limits and coolant	3.3.1.6	4.6.1.7	X		X
Normal operating temperatures	3.3.1.7	4.6.1.8	X	X	
Air inlet/exhaust	3.3.1.8	4.6.1.7	X		X
Oil	3.3.1.9	4.6.1.7	X	X	
Fuel inlet restriction	3.3.1.10	4.6.1.7	X		X
Slope operation	3.3.1.11	4.6.1.9	X		X
Endurance	3.3.1.12	4.6.1.10	X		X
Steam and waterjet cleaning	3.3.2	4.6.2	X		X
Leakage	3.3.3	4.6.3	X	X	X
Environmental	3.3.4	4.6.4	X		X
Electromagnetic radiation	3.3.5	4.6.5	X		X

4.6.1.6 Engine starting. To determine conformance to 3.3.1.5, verify that the engine starts in not more than one minute in an ambient temperature range of minus 25°F (-32°C) to plus 120°F (49°C).

4.6.1.7 Operating limits. During performance capability tests and endurances capability test (see 3.3), operating temperatures, air inlet/exhaust restrictions and fuel inlet restriction shall not exceed the values specified in 3.3.1.7, 3.3.1.8, 3.3.1.10, respectively. Oil and fuel shall be in accordance with 3.3.1.9 and 3.3.1.10.1 and coolant shall conform to specifications given in 3.3.1.6.

4.6.1.8 Normal operating temperatures. Measure the oil sump temperature (at the drain plug) and coolant temperature (at the inlet to the transmision oil cooler) to verify that the requirements of 3.3.1.7 are obtained.

4.6.1.9 Slope operation. Verify that the engine will start and operate on the slopes specified in paragraph 3.3.1.11 and assure that no fuel leakage, coolant leakage or inadequate lubrication occurs.

4.6.1.10 Endurance. Verify that the engine has passed the AEP-5, NATO standard engine 400 hour endurance laboratory test specified in 3.3.1.12.

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4.6.2 Steam and waterjet cleaning. Verify that no deterioration or leakage of seals, gaskets, and electrical components occurs following completion of paragraph 3.3.2 requirements.

4.6.3 Leakage. To determine conformance to 3.3.3, there shall be no evidence of "leaks" or "drips" (see 6.3) during or after testing as specified herein.

4.6.4 Environmental conditions. To determine conformance to 3.3.4 the engine shall be tested in accordance with MIL-STD-810, method 501.1, procedure II, and method 502.1, procedure I.

4.6.5 Electromagnetic radiation. To determine conformance to 3.3.5 the engine shall be tested to meet the functional suppression requirements of MIL-STD-461.

4.7 Protective finish. Verify that the engine meets the requirements of 3.4.

4.8 Identification and marking. The assembly shall be visually examined to verify conformance to the marking requirements of 3.5.

4.8.1 Nameplate. As required by 3.5.1, verify that nameplates, data, and instruction plates shall conform to MIL-P-514.

4.9 Workmanship. Visual examinations shall be performed at all phases of fabrication, assembly, and tests to verify conformance to 3.6.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the packaging data sheet for the desired level of protection specified (see 6.2).

6. NOTES

6.1 Intended use. The diesel engine covered by this specification is intended for use as original equipment, as spares or as replacement in the Armored Combat Earthmover M9 ACE.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Drawing number, title, and revision date.
c. Preproduction inspection, if required (see 4.4.1).
d. Initial production inspection, if not required (see 4.4.2).
e. Selection of applicable levels of preservation, packaging, packing and marking (see 5.1).

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6.3 Definitions.

6.3.1 Leakage. The following definitions are established as guidelines governing leakage:

a. Weep - Any evidence of fluid beyond a seal or joint.
b. Seep - Any evidence of fluid beyond a seal or joint that does not result in the formation of a droplet.
c. Leak - Any evidence of fluid beyond a seal or joint that results in the formation of a droplet.
d. Drip - Any evidence of fluid beyond a seal or joint where droplets form and fall.

6.4 Item definition. The Cummins Model V903 diesel engine is a liquid cooled, 4 cycle, 90° vee, 8 cylinder internal combustion, compression ignition engine. At 2600 revolution per minute (rpm), the V903 diesel engine is rated at 295 horsepower (hp) (220 kw) per SAE Standard J1349 rating conditions. In this specification the diesel engine, and the associated accessories furnished by the engine supplier, are referred to as "engine".

6.5 Painting practices. When manufacturer's commercial practices are to be used, the manufacturer's proposal will specify the method for cleaning, treating and painting, for Government evaluation (see 3.4).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

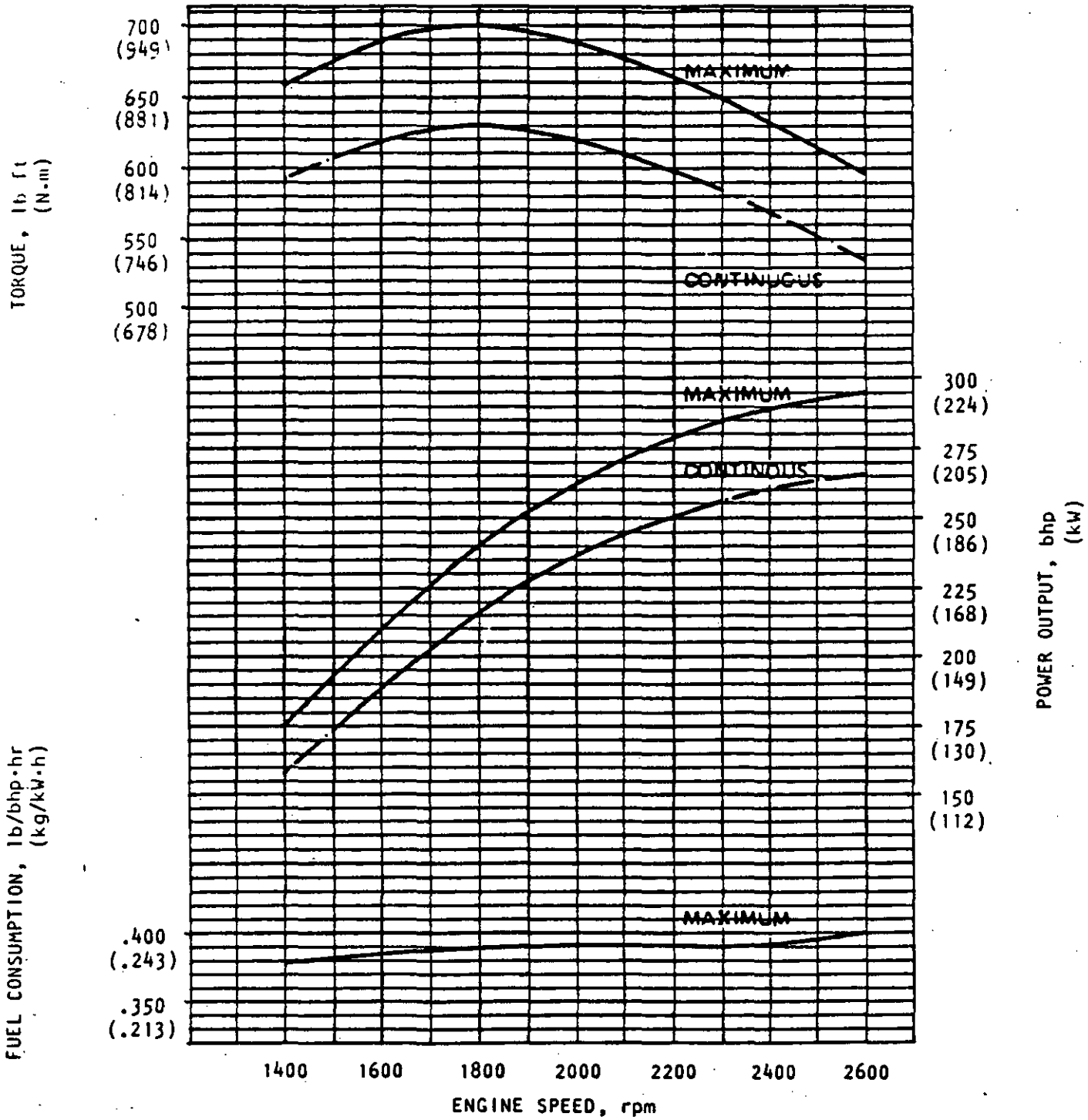
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All data is based on the engine operating with fuel system, water pump, lubricating oil pump, air cleaner, and muffler; not included are alternator, compressor, fan, optional equipment and driven components.



Curves shown above represent engine performance capabilities at SAE standard J1349 conditions of 500 ft (150 m) altitude [29.00" Hg (736 mm Hg) dry barometer], 85°F (29°C) air intake temperature, and 0.38" Hg (9.6 mm Hg) water vapor pressure with No. 2 diesel fuel.

FIGURE 1. Performance data.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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